

201

Next largest element :- (nearest greater to right).

arr = 1 3 2 4
 ↑ ↑ ↑ ↑
 3 4 4 1 → Output.

arr = 1 3 0 0 1 2 4
 ↑ ↑ ↑ ↑ ↑ ↑ ↑
 3 4 2 1 2 4 -1 → output

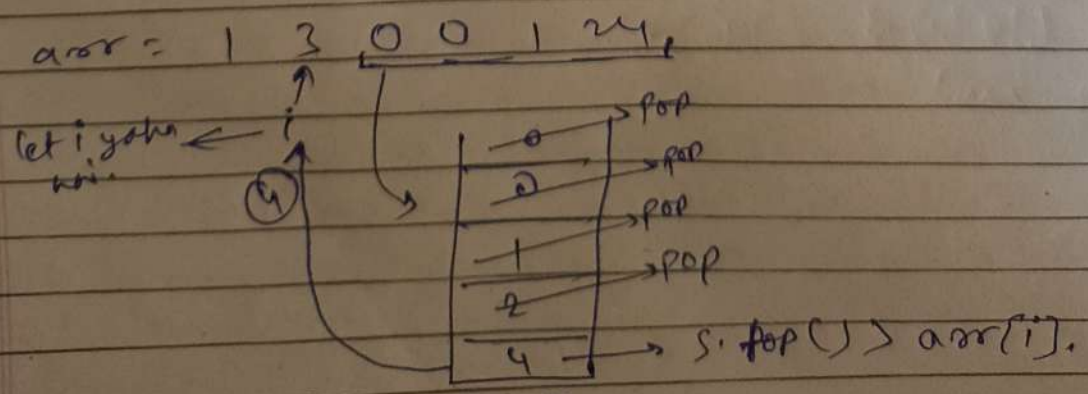
Is ek element ko fix karo aur uske right me fix element se just greater kon hai.

To brute force.

```
for(i=0; i<n; i++) {
  for(j=i+1; j<n; j++) {
  }
```

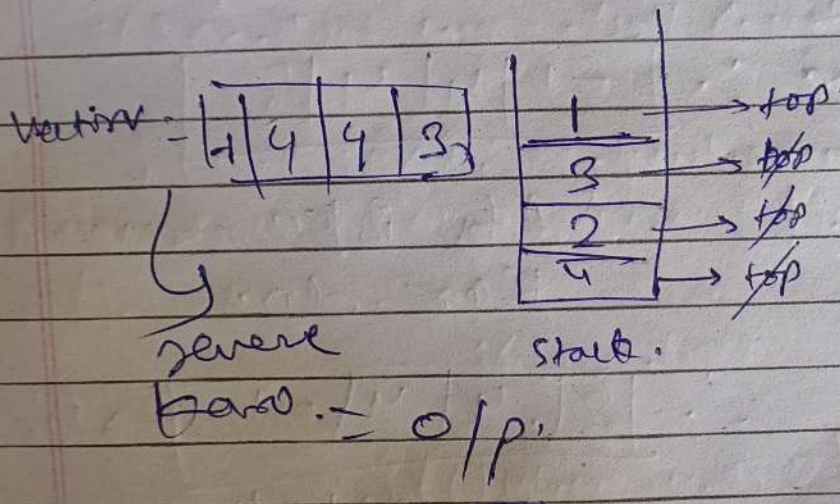
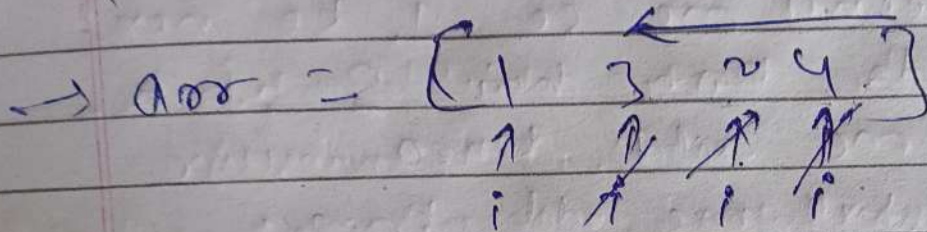
↳ Yaha i, j pe depend kar raha hai, to hme start ke baad me sochna hai.

Now,



Stack.
A LIFO

→ let agar 4 ke jagah 3 se bhi chhota element hota toh stack empty ho jata aur tab hm -1 return krte.



i.e.
→ from right side. 4 ke next bhi element hai, nhi hai toh -1 vector me store krna aur stack ke top me 4 ko store krna. then $i--$.

→ Ab $i=2$, i ke right me 4 hai, jo 2 se bada hai toh vector me 4 store krna aur stack me ~~2~~ store kr top me 2 store, $i--$

→ Ab $i=3$, i ke right me 2 & 4 hai, toh dono me se 3 se bada hai 4, toh vector me 4 store

Karo aur stack me 3 ko add
Karo aur top point kar do, i.e.

→ Ab $i=1$, i ke right me 3, 2, 4
hai, toh teeno me se 1 ke next
me use bada kon hai 3, toh
3 ko vector me add karo aur
us me ko stack me add karo.

Final vector :- $[-1 | 4 | 4 | 3]$

Ab isko reverse
kar do.

→ $[3 | 4 | 4 | -1]$ → Final
output

Quick Summary:-

→ Stack Empty → -1

→ $s.top() > arr[i]$ → $s.top()$

→ $s.top() <= arr[i]$ → pop

- stack empty
- $s.top()$ greater
than $arr[i]$

// code :-

```
vector<int> v;  
stack<int> stack;
```

```
for (int i = size-1; i >= 0; i--) {  
    if (s.size() >= 0) {  
        v.push_back(s.top());  
    }  
    else if (s.size() > 0 && s.top() > arr[i]) {  
        v.push_back(s.top());  
    }  
    else if (s.size() > 0 && s.top() <= arr[i]) {  
        while (s.size() > 0 && s.top() <= arr[i]) {  
            s.pop();  
        }  
        if (s.size() >= 0) {  
            v.push_back(s.top());  
        }  
        else {  
            v.push_back(s.top());  
        }  
        s.push(arr[i]);  
    }  
}
```


~~Class Soln~~

~~Public static void main String~~

~~class Soln~~ {

~~Public int[] nextGreaterElements (int[]~~
~~num1, int[] num2) {~~

~~HashMap<Integer, Integer> map = new HashMap<>();~~
~~Stack<Integer> stack = new Stack<>();~~

~~for (int i = num1.length - 1; i > 0; i--) {~~
~~while (!stack.isEmpty() & & stack.peek()~~
~~<= num2[i]) {~~
~~stack.pop();~~

~~if (stack.isEmpty()) {~~
~~map.put(num2[i], -1);~~

~~} else {~~
~~map.put(num2[i], stack.peek());~~

~~stack.push(num2[i]);~~

~~int[] res = new int[num1.length];~~
~~for (int i = 0; i < num1.length; i++) {~~
~~res[i] = map.get(num1[i]);~~

~~return res;~~

~~}~~